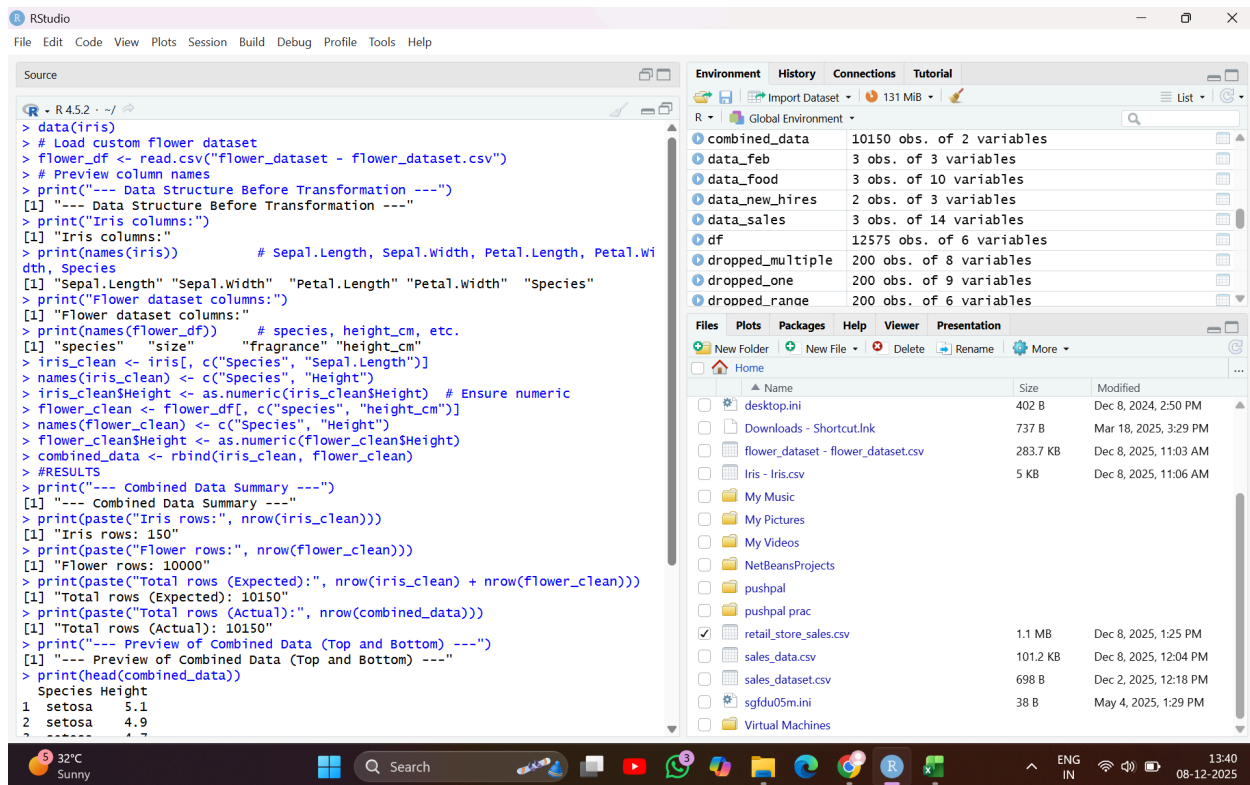


SHETH L.U.J AND SIR M.VCOLLEGE

Subject: Data Analysis with SAS / SPSS /R

Practical no. 12

Aim:Combining datasets vertically
(concatenation) using rbind() (R).



The screenshot shows the RStudio interface. The Source pane on the left contains R code that loads two datasets, 'iris' and 'flower_dataset.csv', cleans them, and combines them into 'combined_data' using the `rbind()` function. The Environment pane on the right shows the objects in the Global Environment, including 'combined_data' (10150 obs. of 2 variables), 'data_feb' (3 obs. of 3 variables), 'data_food' (3 obs. of 10 variables), 'data_new_hires' (2 obs. of 3 variables), 'data_sales' (3 obs. of 14 variables), 'df' (12575 obs. of 6 variables), 'dropped_multiple' (200 obs. of 8 variables), 'dropped_one' (200 obs. of 9 variables), and 'dropped_range' (200 obs. of 6 variables). The Files pane at the bottom shows the file explorer with various files and folders.

```
> data(iris)
> # Load custom flower dataset
> flower_df <- read.csv("flower_dataset - flower_dataset.csv")
> # Preview column names
> print("---- Data Structure Before Transformation ----")
> print("Iris columns:")
[1] "Iris columns:"
> print(names(iris))      # Sepal.Length, Sepal.Width, Petal.Length, Petal.Wi
dth, Species
[1] "Sepal.Length" "Sepal.Width" "Petal.Length" "Petal.Width" "Species"
> print("Flower dataset columns:")
[1] "Flower dataset columns:"
> print(names(flower_df)) # species, height_cm, etc.
[1] "species" "size"      "fragrance" "height_cm"
> iris_clean <- iris[, c("Species", "Sepal.Length")]
> names(iris_clean) <- c("Species", "Height")
> iris_clean$Height <- as.numeric(iris_clean$Height) # Ensure numeric
> flower_clean <- flower_df[, c("species", "height_cm")]
> names(flower_clean) <- c("Species", "Height")
> flower_clean$Height <- as.numeric(flower_clean$Height)
> combined_data <- rbind(iris_clean, flower_clean)
> #RESULTS
> print("---- Combined Data Summary ----")
[1] "---- Combined Data Summary ----"
> print(paste("Iris rows:", nrow(iris_clean)))
[1] "Iris rows: 150"
> print(paste("Flower rows:", nrow(flower_clean)))
[1] "Flower rows: 10000"
> print(paste("Total rows (Expected):", nrow(iris_clean) + nrow(flower_clean)))
[1] "Total rows (Expected): 10150"
> print(paste("Total rows (Actual):", nrow(combined_data)))
[1] "Total rows (Actual): 10150"
> print("---- Preview of Combined Data (Top and Bottom) ----")
[1] "---- Preview of Combined Data (Top and Bottom) ----"
> print(head(combined_data))
  Species Height
1  setosa   5.1
2  setosa   4.9
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```

R - R4.5.2 - ~/
[1] "species" "size" "fragrance" "height_cm"
> iris_clean <- iris[, c("Species", "Sepal.Length")]
> names(iris_clean) <- c("Species", "Height")
> iris_clean$Height <- as.numeric(iris_clean$Height) # Ensure numeric
> flower_clean <- flower_df[, c("species", "height_cm")]
> names(flower_clean) <- c("Species", "Height")
> flower_clean$Height <- as.numeric(flower_clean$Height)
> combined_data <- rbind(iris_clean, flower_clean)
> #RESULTS
> print("--- Combined Data Summary ---")
[1] "--- Combined Data Summary ---"
> print(paste("Iris rows:", nrow(iris_clean)))
[1] "Iris rows: 150"
> print(paste("Flower rows:", nrow(flower_clean)))
[1] "Flower rows: 10000"
> print(paste("Total rows (Expected):", nrow(iris_clean) + nrow(flower_clean)))
[1] "Total rows (Expected): 10150"
> print(paste("Total rows (Actual):", nrow(combined_data)))
[1] "Total rows (Actual): 10150"
> print("--- Preview of Combined Data (Top and Bottom) ---")
[1] "--- Preview of Combined Data (Top and Bottom) ---"
> print(head(combined_data))
  Species Height
1  setosa    5.1
2  setosa    4.9
3  setosa    4.7
4  setosa    4.6
5  setosa    5.0
6  setosa    5.4
> print(tail(combined_data))
  Species Height
10145   rose    87.69
10146 hibiscus 109.52
10147 shoeblack plant 145.23
10148   hibiscus 126.69
10149 shoeblack plant  77.62
10150   rose    88.11
>
>

```

Environment History Connections Tutorial

R - Global Environment

- combined_data 10150 obs. of 2 variables
- data_feb 3 obs. of 3 variables
- data_food 3 obs. of 10 variables
- data_new_hires 2 obs. of 3 variables
- data_sales 3 obs. of 14 variables
- df 12575 obs. of 6 variables
- dropped_multiple 200 obs. of 8 variables
- dropped_one 200 obs. of 9 variables
- dropped_range 200 obs. of 6 variables

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

Home

Name	Size	Modified
desktop.ini	402 B	Dec 8, 2024, 2:50 PM
Downloads - Shortcut.lnk	737 B	Mar 18, 2025, 3:29 PM
flower_dataset - flower_dataset.csv	283.7 KB	Dec 8, 2025, 11:03 AM
Iris - Iris.csv	5 KB	Dec 8, 2025, 11:06 AM
My Music		
My Pictures		
My Videos		
NetBeansProjects		
pushpal		
pushpal prac		
retail_store_sales.csv	1.1 MB	Dec 8, 2025, 1:25 PM
sales_data.csv	101.2 KB	Dec 8, 2025, 12:04 PM
sales_dataset.csv	698 B	Dec 2, 2025, 12:18 PM
sgfdu05m.ini	38 B	May 4, 2025, 1:29 PM
Virtual Machines		

32°C Sunny

Search

ENG IN

13:40 08-12-2025